v5 layerwise self-distillation — joint campaign report

Gemma-4-31B · 2071-item corpus (Machado + Quijote) · runs of 2026-07-26 → 2026-08-15 · report built 2026-08-16

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LAW  Teacher = same model, adapters off, WITH passage (one no-grad pass records per-layer hidden targets h_t[L] at answer
rows). Student = model + LoRA with the passage REMOVED. One forward; every block input detached by hook, so block L
transforms its own censored trajectory h_s[L-1] and its local loss distance(y_L, h_t[L]) roots only in block L's LoRA.
Output KL/CE vs teacher are EVALUATION ONLY (optimizer weight zero); embeddings / final norm / unembedding frozen.

BASELINES (epoch 0, untrained, censored prompts)   mach recall 0.171-0.173 · quij 0.178 · arc_easy 0.32 · argmax 0.421

CAMPAIGN VERDICTS
  1. The metric is the first-order lever: huber destroyed the model by e1-e8; vocab_mse found the non-destructive band.
  2. Aligned positions beat natural (baseline crossing vs none).
  3. Every dense-writing recipe (layer_gate 'all') erodes to the destruction auto-abort, at ANY rank (r8/r32/r128)
     and any LR (1e-5, 3e-5); r128 erodes worst, lr 3e-5 roughly doubles erosion speed.  Rank + LR axes CLOSED.
  4. Only extreme write-sparsity survives 300 epochs (topk_abs:1 = ~1 layer written/step).
  5. Best arm: delta_cosine + topk_abs:1 + aligned (LONG, job 431670): recall above baseline in 19/30 evals,
     plateau 0.18-0.19 over e60-e230, peak 0.1921 @e160/170, fading back to baseline by e240-e300.
  6. Mechanism: the increment (delta_cosine) gate starts mid-stack (depth shares 14/52/34% over e1-40, exactly
     replicating the 40-epoch arm) then COLLAPSES onto L54, which takes ~84% of all selections from e81 on.
     The increment metric moves the attractor (L54 vs vocab_mse's L58) but does not prevent collapse; the
     above-baseline recall plateau coincides with the L54-concentrated phase.
  7. Deployment artifacts: every eval epoch is checkpointed; best-checkpoint rule (max recall s.t. argmax >= 0.8 x e0)
     selects v5p_dcos_topkabs1_long/checkpoint_e160 and _e170.

OPEN QUESTIONS FOR THE NEXT CAMPAIGN
  (a) Does a SCHEDULED mid-stack distribution (per-layer quotas / epoch-frozen thresholds) beat emergent L54-only?
  (b) Checkpoint-selection deployment; (c) the vN attention-top-k censor.

v4 CLOSURE (context, details on the last page): neither loss, LR, rank, nor KV-refresh makes the teacher-frozen
block-local law compose into behavior change. v5's self-trajectory law is the only line that learned.
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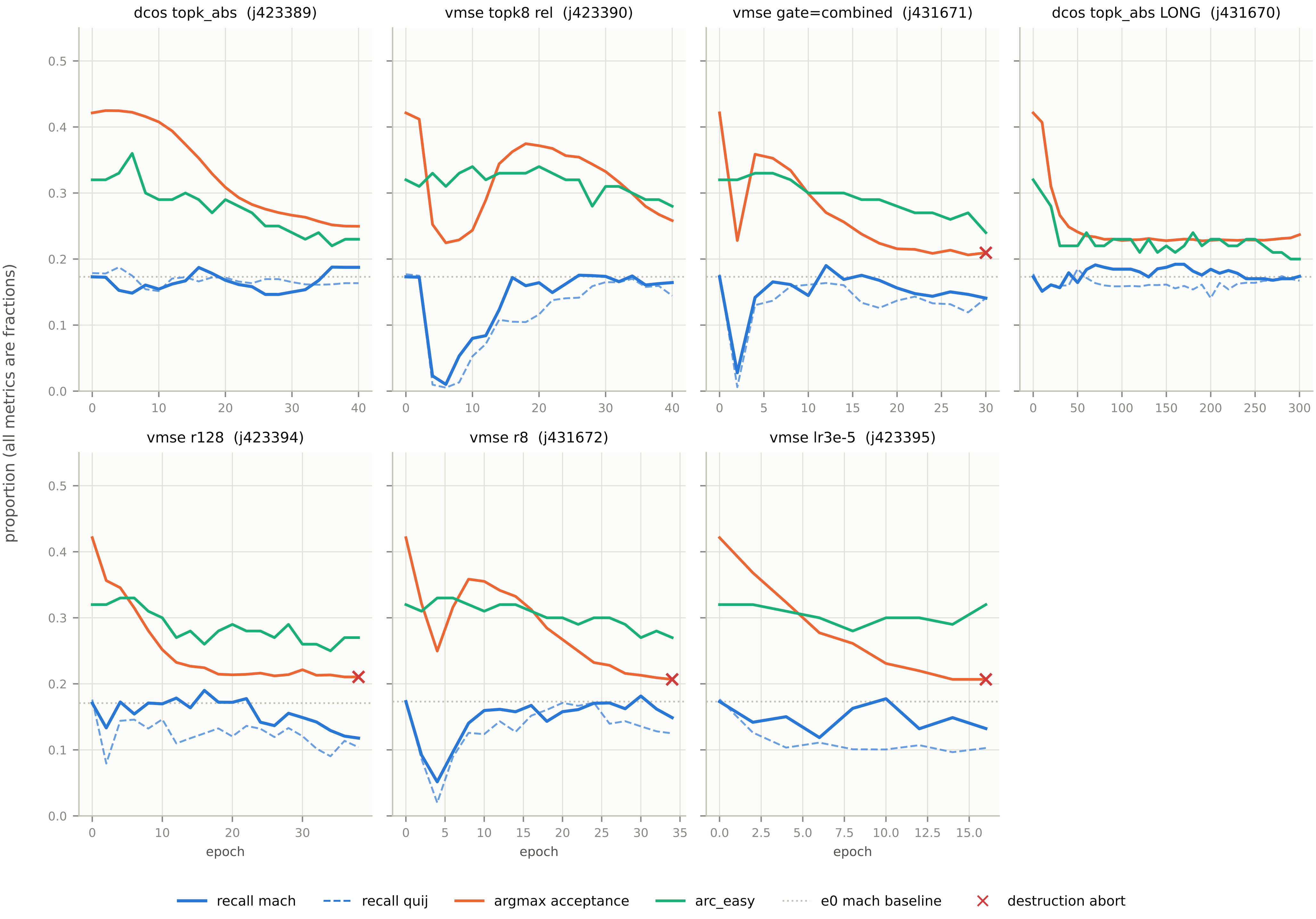
All v5 runs — configuration and fate

run	job	loss	gate	r	lr	pos	epochs	fate	peak mach	>base	final argmax	final arc
trainv5_g31b_selfdistill_lr1e4_de	423270	huber	all	32	1e-04	?	8	e5	0.0000@e1	0	0.0000	0.26
		note: destroyed e1; L59 contaminated										
trainv5_g31b_selfdistill_huber_an	423293	huber	all	32	1e-05	?	40	e8	0.0590@e2	0	0.0000	0.19
		note: destroyed e4-8; L59 contaminated										
trainv5_g31b_vmse_l59bug_partial	423303	vocab_mse	all	32	1e-05	?	40	e2	0.0000@e2	0	0.0000	0.18
		note: superseded										
trainv5_g31b_vmse	423314	vocab_mse	all	32	1e-05	natural	40	ABORT e26	0.1759@e14	1	0.2132	0.27
trainv5_g31b_vmse_aligned	423325	vocab_mse	all	32	1e-05	aligned	40	ABORT e28	0.1903@e13	7	0.2072	0.26
trainv5_g31b_vmse_al_clip001	423340	vocab_mse	all	32	1e-05	aligned	30	done e30	0.1950@e6	5	0.2121	0.23
trainv5_g31b_vmse_al_sema1	423350	vocab_mse	surprise_em	32	1e-05	aligned	30	done e30	0.1943@e30	4	0.2202	0.28
v5p_topkabs1_long	423387	vocab_mse	topk_abs:1	32	1e-05	aligned	300	done e300	0.1755@e150	3	0.4053	0.29
v5p_surprise_long	423388	vocab_mse	surprise_em	32	1e-05	aligned	300	ABORT e60	0.1795@e30	2	0.2086	0.21
v5p_dcos_topkabs1	423389	delta_cosin	topk_abs:1	32	1e-05	aligned	40	done e40	0.1878@e36	5	0.2497	0.23
v5p_topk8_rel	423390	vocab_mse	topk:8	32	1e-05	aligned	40	done e40	0.1755@e26	4	0.2582	0.28
v5p_signal_combined	431671	vocab_mse	all	32	1e-05	aligned	40	ABORT e30	0.1901@e12	2	0.2095	0.24
v5p_dcos_topkabs1_long	431670	delta_cosin	topk_abs:1	32	1e-05	aligned	300	done e300	0.1921@e160	19	0.2369	0.20
		note: concluding arm										
v5p_r128	423394	vocab_mse	all	128	1e-05	aligned	40	ABORT e38	0.1900@e16	7	0.2104	0.27
v5p_r8	431672	vocab_mse	all	8	1e-05	aligned	40	ABORT e34	0.1815@e30	1	0.2067	0.27
v5p_lr3e5	423395	vocab_mse	all	32	3e-05	aligned	40	ABORT e16	0.1774@e10	1	0.2067	0.32

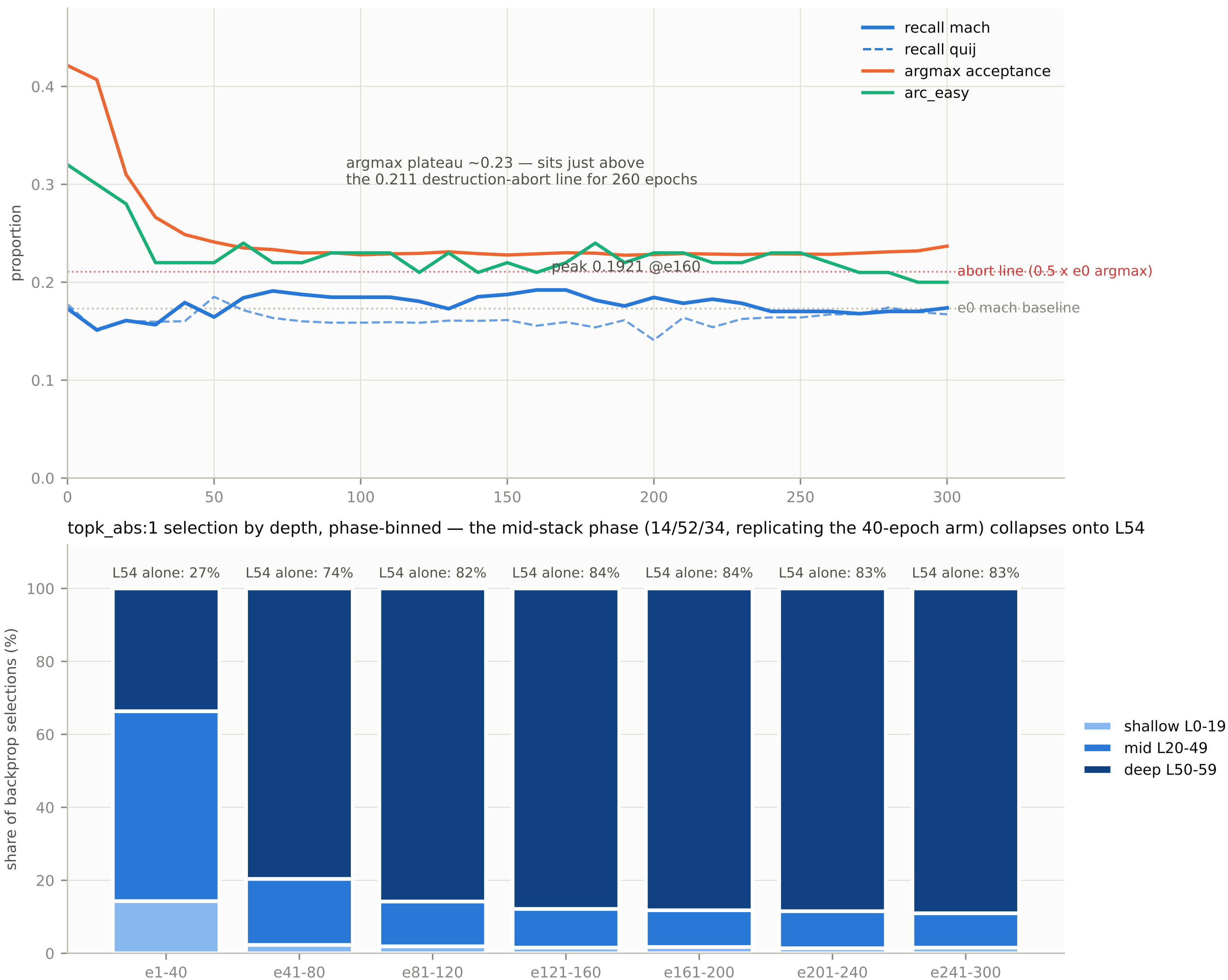
peak mach = best Machado recall after epoch 0 (word-LCS vs teacher answer, fixed 24-sample subset; each run's own e0 is its baseline).
>base = number of evals above the run's epoch-0 baseline. ABORT = destruction auto-abort (argmax < 0.5 x e0 at two consecutive evals).
attempt1/attempt2 predate the hidden_states[-1] fix (f35f74a): their L59 surprise values are contaminated; L0-58 valid.

Run curves (1/2) — destruction era and the vocab_mse band

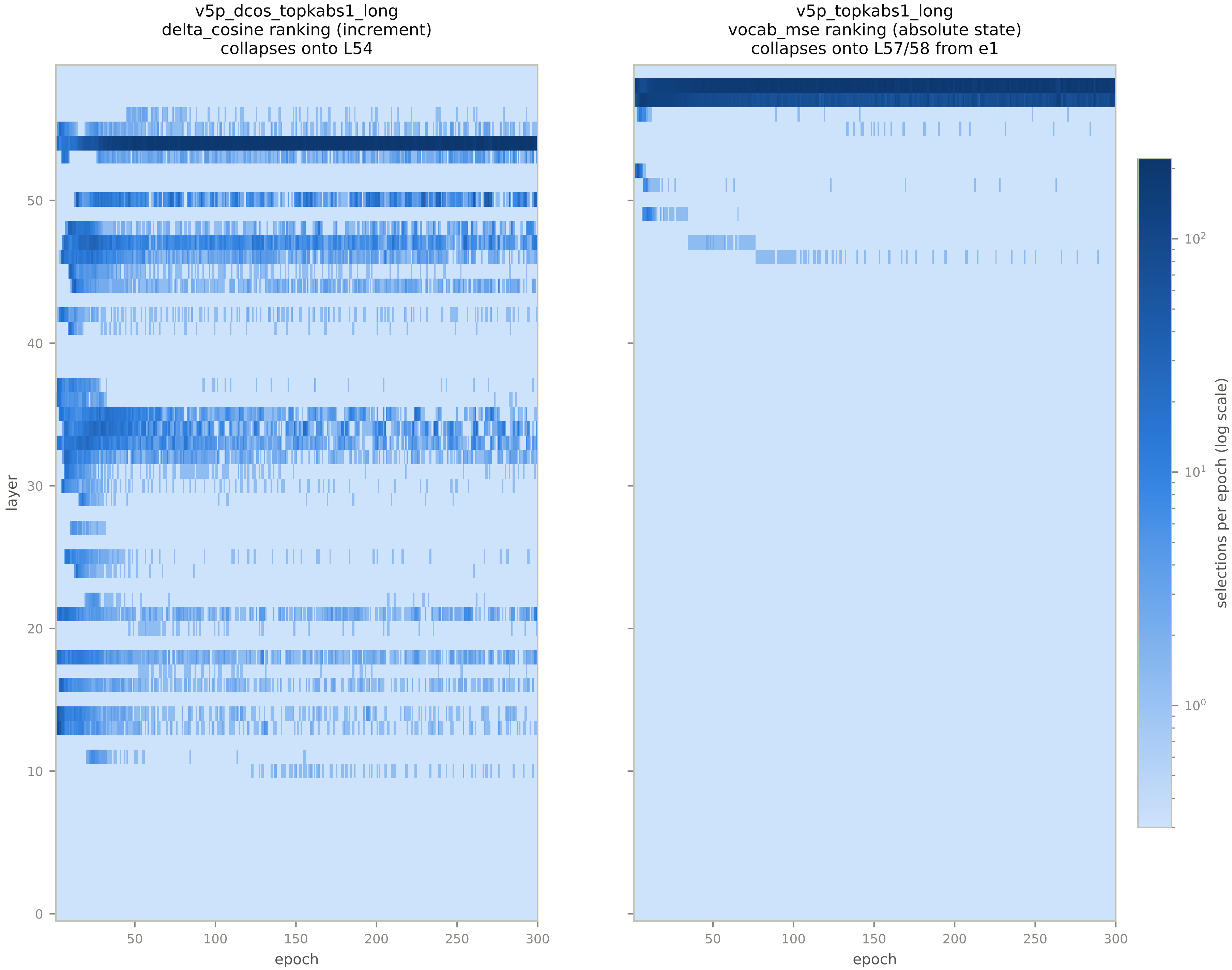
Run curves (2/2) — paced campaign arms



Concluding arm — v5p_dcos_topkabs1_long (delta_cosine + topk_abs:1 + aligned, 300 ep, j431670)

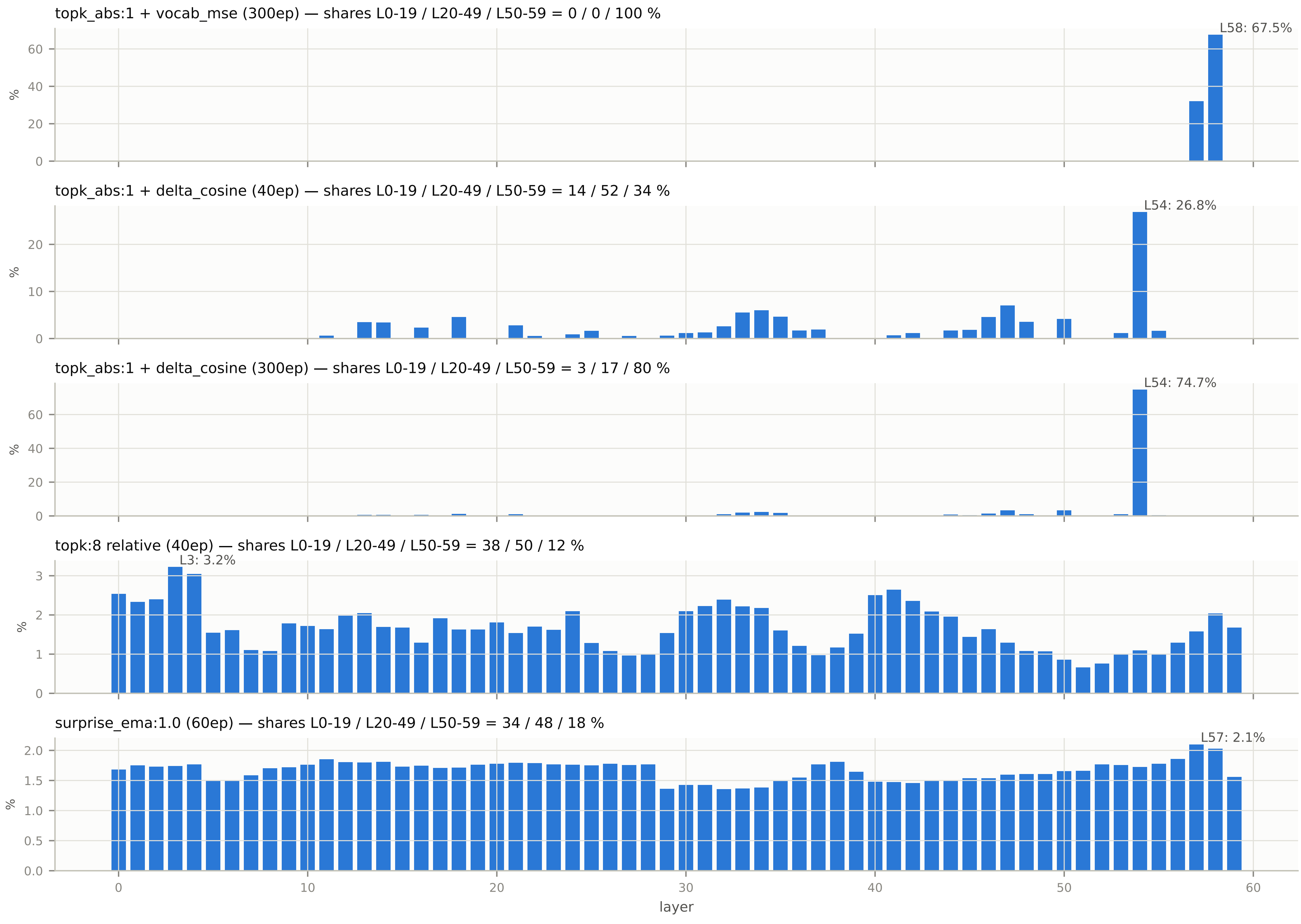


Where the absolute gate writes — backprop selections per layer x epoch (log color)

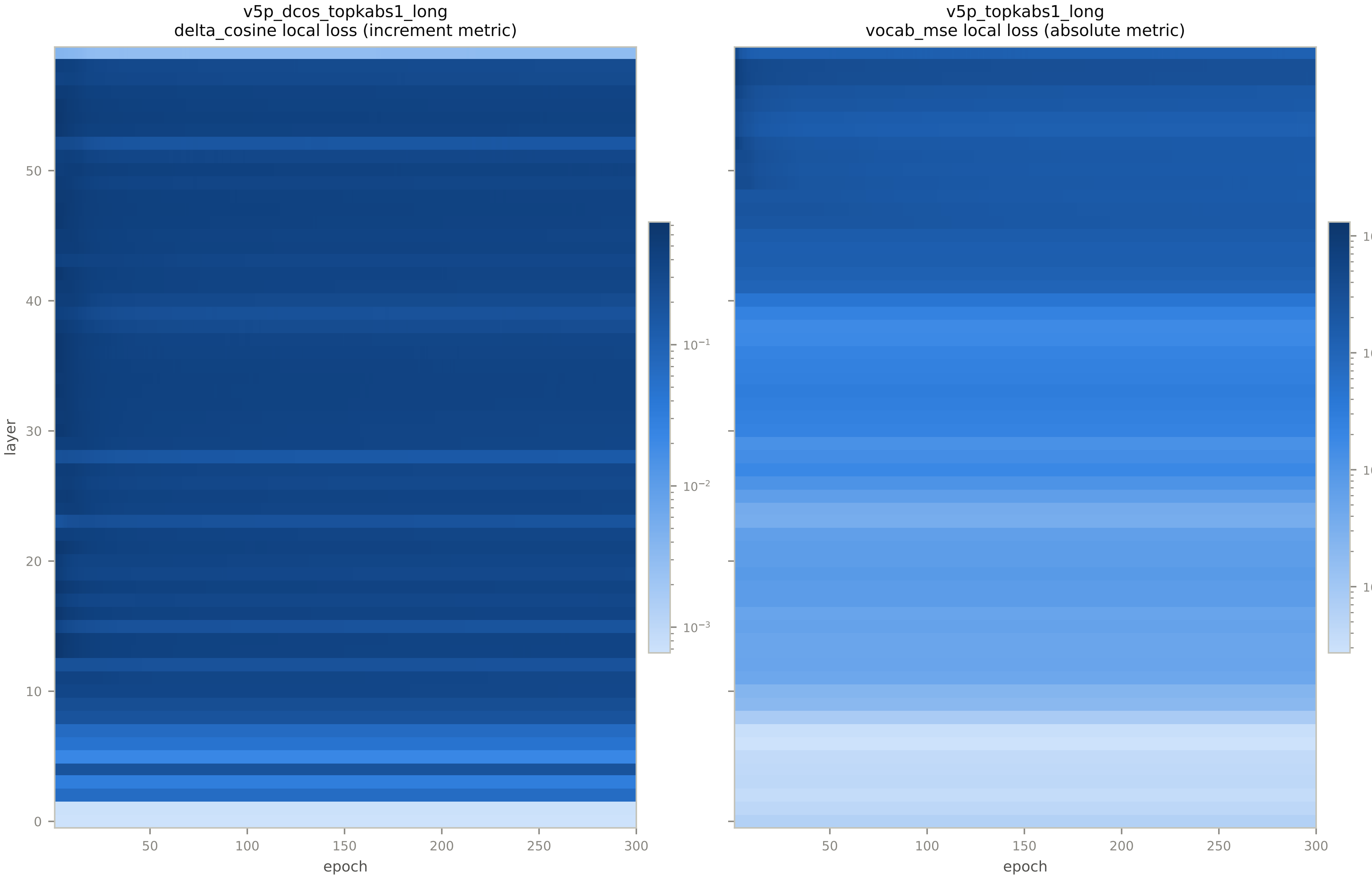


Both gates backprop ~1 layer per step; color shows which layer won each step's absolute-loss ranking, per epoch.

Aggregate backprop share per layer — every gated arm

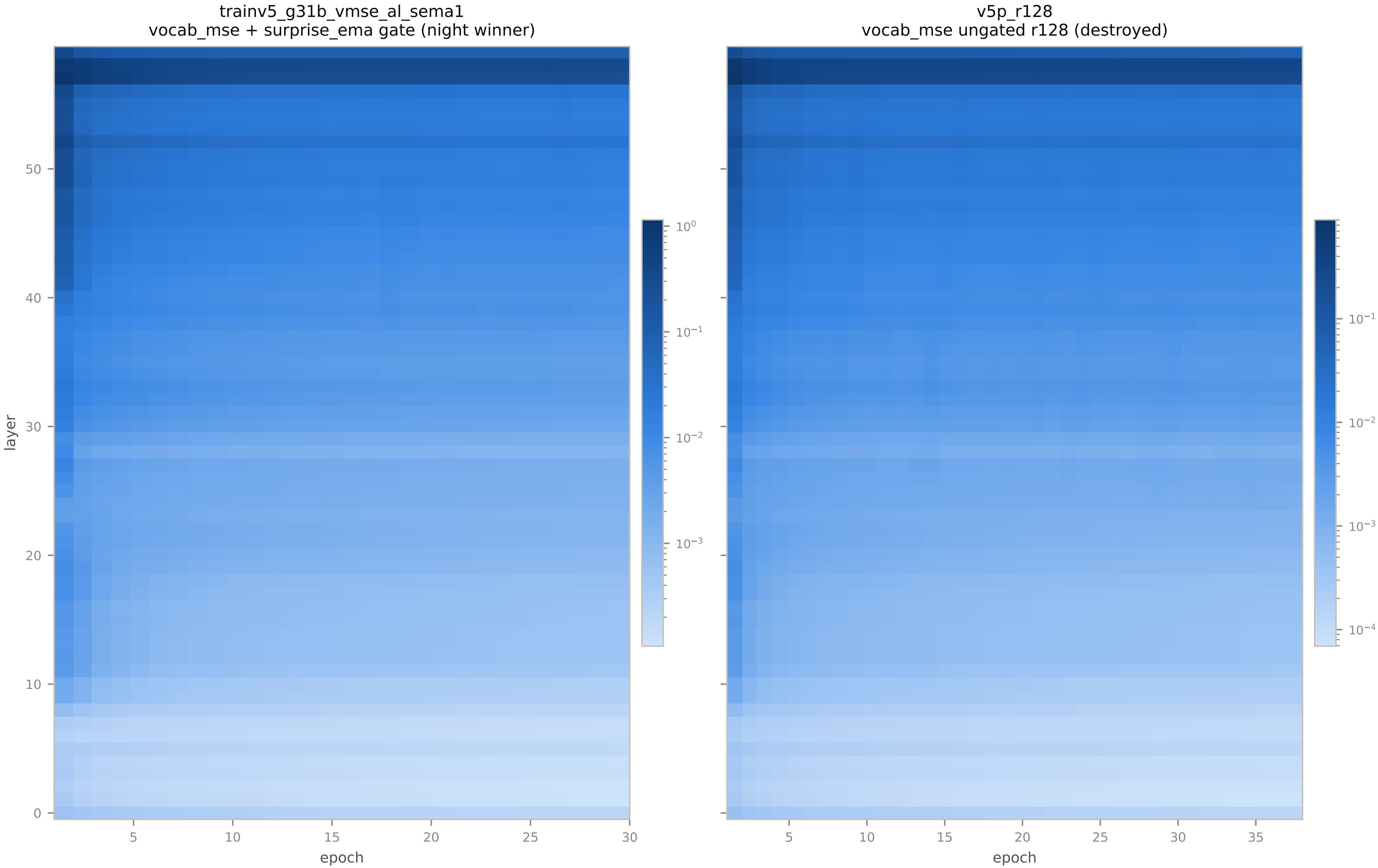


Surprise profiles — per-layer local loss of the censored student vs teacher h_t[L], per epoch (log color)



Profile-reading rule (owner, 2026-07-27): a profile = monotone drift envelope + span-periodic attention component (Gemma-4 full-attention layers 5,11,...,59) + residual; only the residual localizes memorization. Loss units differ per metric — compare shapes within a panel, not magnitudes across panels.

Surprise profiles — per-layer local loss of the censored student vs teacher h_t[L], per epoch (log color)



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Context: the v4 line is closed (negative on every axis)

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The v5 campaign exists because v4 (teacher h[L-1] -> owned block L -> teacher h[L], frozen teacher attention context)
never produced behavior change. Each axis was tested and closed:

  A. RANK (r64 campaign, all models, jobs 422979-996) ......... CLOSED: dense arms bit-flat (q27b argmax 0.5560->0.5561,
     g31b 0.4678->0.4678); MoE r64 actively harmed (q35b huber 0.5433->0.4445). Rank was never the constraint.
     Side-datum: the A4B SP-vs-shard numerics discrepancy (4.0e-7) is RANK-driven — r16+experts passed twice at ~4e-8.

  B. LOSS x LR (q27b 2x2 screen, jobs 423027-032) ............. CLOSED: none moved. vocab_mse@3e-6 flat; huber@3e-5
     degraded. Block-local training with teacher-frozen context cannot compose, whatever the loss or step size.

  C. KV-STALENESS (A4B refresh pair, jobs 429743/429745) ...... CLOSED: adapter-refreshed K/V indistinguishable from
     teacher_frozen over ~230 epochs (argmax 0.45->0.36 both, recall 0.00 both). Hypothesis [393] falsified.

  D. WHY: the v4 block-local objective is near-trivial — each block is fed the exact teacher context, so hidden loss
     starts near zero and provides no useful gradient. v5 removes the passage from the student, making the local
     residual (censored-self vs passage-informed teacher) large at every layer: the objective v4 never had.

CAMPAIGN TIMELINE
  Jul 25-26   v4 closure runs; trainv5.py written (monolith, layerwise-certified at runtime)
  Jul 26      attempt 1 (huber, lr 1e-4): destroyed in 1 epoch.  attempt 2 (huber + self-anchor, lr 1e-5): destroyed by e4-8
  Jul 27      the band found: vocab_mse + aligned; surprise_ema healthiest (0.194 rising @e30); two-week schedule installed
  Aug 1       self-review #1: vocab_mse topk_abs = 99.8% tail concentration, stable but weak
  Aug 6       self-review #2: increment thesis confirmed — dcos ranking redistributes mid-stack (14/52/34), best arm
  Aug 12      self-review #3: C-line closed; corrupted node venv fixed (3 arms lost 5s after launch, all recovered)
  Aug 12-15   concluding runs: dcos long (300ep, completed), signal_combined / r8 / r128 / lr3e5 (all destruction-aborted)
  Aug 16      this report; campaign concluded

All numbers in this report are read from each run's metrics.jsonl as emitted by the trainer; evaluation metrics
(recall, arc_easy, argmax acceptance, KL/CE_eval) carry optimizer weight zero by construction (frozen-vocabulary law).
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